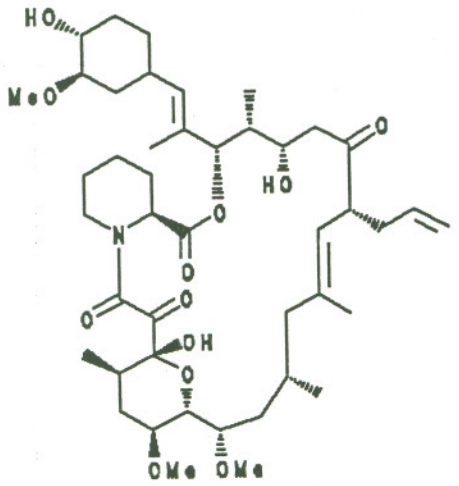
ORGANIC CUMULATIVE EXAMINATION
April 28th 1990

Introduction

In recent years the macrocyclic lactone FK-506 has been at the center of much attention. This species has been shown to exhibit some remarkable immunosuppressant properties, along with a number of other physiological properties. This fascinating compound has been shown to be active at concentrations 100 times lower than cyclosporin, which is one of the more commenly prescribed immunosuppressant drugs.

One unique feature of the structure of FK-506 is the tricarbonyl array (amide through hemiacetal). It is believed that this array is partially the source of the biological activity of this species. Thus much attention has focused upon the synthesis of this tricarbonyl array.

Ref : A.V. Rama Rao, T.K. Chakraborty, K. Laxma Reddy Tetrahedron Letters 1990, 31, 1439.



FK-506

Figure 1

Questions :

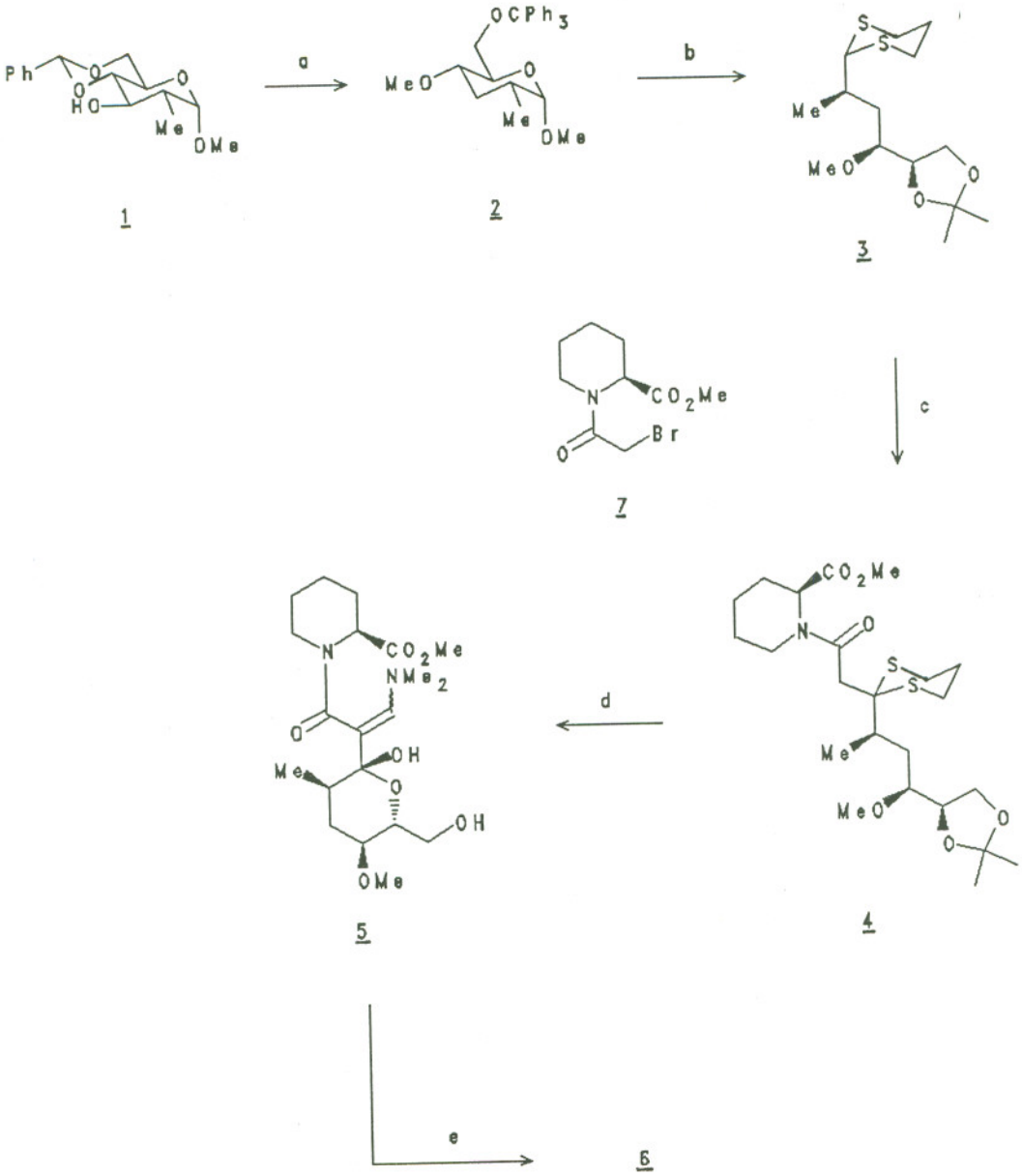
1. Show the mechanism of the four step transformation of 1 to give 2.(40pts)

2. Show the mechanism of the two step transformation of 2 to give 3.(25pts)

3. Draw the mechanism of the formation of 4 from 3.(10pts)

4. Show the mechanism of the multi-step sequence which transforms 4 into 5.(40pts)

5. What is the product of the final reaction, structure 6?(15pts)



Reaction conditions:

a : i) NaH(1.3eq.), CS_2(1.2eq.), Et_2O, reflux, 3hrs, followed by addition of MeI(1.2eq.),
ii) n-Bu_3SnH(1.5eq.), AIBN(cat.), toluene, reflux, 12hrs,
iii) p-toluenesulfonic acid(0.4eq.), MeOH, r.t., 6-7hrs,
iv) Ph_3CCl(1.2eq.), pyridine, 55-60°C, 4hrs,
v) NaH(1.5eq.), MeI(1.5eq.), DMF:THF (3:7), 0°C, 2hrs.

b : i) 1,3-propanedithiol(1.1eq.), $BF_3.Et_2O$(1.0eq.), CH_2Cl_2, 0°C, 2hrs,
ii) $Me_2C(OMe)_2$, p-toluenesulfonic acid(0.2eq.), 12hrs.

c : n-BuLi, THF, -15°C, 4hrs, followed by the addition of L-N-(α-bromoacetyl)pipecolic acid methyl ester(7) at -78°C, 6-7hrs.

d : i) $BF_3.Et_2O$(2.0eq), HgO(2.0eq.), 15% aqueous THF, followed by the addition of 4 at 0°C, 4hrs,
ii) $(MeO)_2CHNMe_2$(2.0eq.), 80°C, 2hrs,
iii) p-toluenesulfonic acid(0.2eq.), 10% aqueous MeOH, 55°C, 12hrs.

e : O_3,CH_2Cl_2, -78°C, 5-10mins.